

LAAYOUNE, Western Sahara

WMO: 600330

Lat: 27.152N

Lon: 13.219W

Elev: 207

StdP: 14.59

Time Zone: 1.00 (E01)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	50.3	53.1	17.4	13.3	69.5	21.6	16.4	70.2	31.2	66.0	27.2	66.5	7.2	120	0.578

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	18.1	97.0	69.2	92.8	69.0	88.2	68.5	74.7	87.8	73.2	85.8	72.1	83.1	15.0	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.5	117.3	79.4	69.9	111.0	76.3	68.4	105.1	74.9	38.4	87.8	36.9	86.4	35.9	83.6	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.3	27.0	25.0	DB	45.7	107.6	3.4	5.2	43.2	111.3	41.3	114.4	39.4	117.3	36.9	121.0	(4)
(5)			WB	39.1	76.8	2.6	1.8	37.3	78.0	35.8	79.1	34.3	80.1	32.4	81.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	70.7	63.2	64.4	67.6	68.3	70.1	73.0	76.4	78.3	77.0	74.8	69.5	65.3	(6)
(7)		DBStd	6.33	3.61	4.16	4.74	4.58	3.89	3.15	3.70	3.53	3.09	4.20	4.78	3.31	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	207	79	56	19	7	1	0	0	0	0	0	8	37	(9)
(10)		CDD50	7555	409	403	545	550	624	689	819	878	811	769	584	474	(10)
(11)		CDD65	2286	23	39	99	107	159	239	354	413	361	304	142	46	(11)
(12)		CDH74	14451	182	271	684	646	761	1273	2292	2986	2426	1880	798	252	(12)
(13)		CDH80	4983	24	67	273	265	248	308	804	1228	840	641	251	34	(13)
(14)	Wind	WSAvg	14.1	11.5	12.6	13.0	14.4	15.0	16.5	19.9	18.6	14.2	11.0	10.9	11.4	(14)
(15)	Precipitation	PrecAvg	2.2	0.5	0.4	0.2	0.0	0.0	0.0	0.0	0.0	0.1	0.2	0.6	0.6	(15)
(16)		PrecMax	5.0	2.4	2.4	0.8	0.3	0.2	0.0	0.0	0.3	0.2	0.6	1.0	2.3	(16)
(17)		PrecMin	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	1.4	0.7	0.6	0.2	0.1	0.0	0.0	0.0	0.1	0.0	0.2	0.3	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.7	86.2	94.7	98.8	100.6	95.1	102.3	100.8	97.2	98.7	91.6	84.1	(19)
(20)			MCWB	57.1	58.2	61.9	62.4	66.6	70.8	74.3	75.6	71.6	69.3	63.1	57.5	(20)
(21)		2%	DB	78.8	81.0	87.6	86.4	86.2	86.4	91.6	94.8	91.0	91.5	86.3	80.2	(21)
(22)			MCWB	54.8	58.2	61.5	62.3	65.5	69.3	72.1	73.5	71.6	66.5	61.6	56.8	(22)
(23)		5%	DB	75.3	77.0	81.2	80.2	80.6	82.6	86.3	89.6	87.6	86.1	82.3	76.8	(23)
(24)			MCWB	55.1	58.5	61.0	62.9	65.5	67.9	70.6	72.1	70.3	68.3	62.4	57.2	(24)
(25)		10%	DB	71.8	73.1	75.6	75.6	77.2	80.4	83.9	86.0	84.4	82.5	78.5	73.6	(25)
(26)			MCWB	56.2	58.9	61.1	62.3	64.2	66.9	69.2	71.0	69.4	68.1	62.9	57.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.6	66.1	67.9	67.9	70.1	72.9	75.7	78.2	75.7	74.5	70.5	68.1	(27)
(28)			MCD B	72.0	74.4	81.0	82.1	86.5	87.0	91.5	92.4	87.1	85.5	80.4	73.5	(28)
(29)		2%	WB	62.2	63.9	65.4	65.9	68.0	70.7	73.3	75.2	73.6	72.5	68.9	65.5	(29)
(30)			MCD B	69.2	72.3	76.2	77.4	81.0	82.6	89.0	88.4	84.4	81.8	76.9	70.7	(30)
(31)		5%	WB	60.6	62.5	63.9	64.5	66.5	69.2	71.7	73.4	72.3	71.1	67.4	63.5	(31)
(32)			MCD B	67.4	70.7	73.3	75.2	77.1	79.9	84.0	86.9	82.4	79.2	74.8	69.2	(32)
(33)		10%	WB	59.4	60.9	62.7	63.2	65.2	67.8	70.2	72.1	71.0	70.0	65.8	61.8	(33)
(34)			MCD B	66.2	68.4	71.5	72.7	74.7	77.6	80.9	83.3	80.2	77.6	72.8	67.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	18.4	18.0	18.9	16.9	16.1	16.4	17.2	18.1	17.8	17.4	17.5	18.1	(35)
(36)			MCDBR	23.7	25.0	27.6	24.4	22.1	20.3	21.7	23.8	21.4	23.2	22.9	22.6	(36)
(37)			MCWBR	10.7	10.6	9.8	8.3	7.6	6.7	6.6	7.1	6.5	7.6	9.0	10.2	(37)
(38)		5% WB	MCDBR	17.5	19.3	21.4	20.1	19.5	19.6	20.5	21.5	19.0	18.1	17.6	15.8	(38)
(39)			MCWBR	9.3	9.5	8.9	7.8	7.5	7.1	7.2	7.7	6.8	7.3	8.2	8.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.396	0.428	0.466	0.461	0.451	0.518	0.651	0.640	0.535	0.460	0.407	0.406	(40)	
(41)		taud	2.155	2.058	1.950	1.998	2.055	1.884	1.611	1.655	1.886	2.059	2.185	2.140	(41)	
(42)		Ebn at Noon	265	265	262	267	269	250	218	220	240	252	259	254	(42)	
(43)		Edh at Noon	42	49	58	57	54	64	83	80	62	49	40	41	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1281	1567	1934	2203	2348	2375	2278	2132	1921	1604	1317	1158	(44)	
(45)		RadStd	63	76	70	82	81	68	69	67	94	66	55	62	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page